Rodrigo E. L. Córdova Rosado

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Osage Nation Tribal Enrollment No. 025118

KEYWORDS Active Galactic Nuclei (16), Observational cosmology (1146), Gravitational Lensing (670)

EDUCATION

Princeton University, Princeton, New Jersey, USA

Ph.D. in Astrophysical Sciences — Aug 2020 – May 2025

Dissertation Title: "Black Holes and Dark Matter:
On the Connection between Active Galactic Nuclei and their Halo Environments"
Advisors: Jenny E. Greene & Andy D. Goulding *Defended*: May 27, 2025

M.A. in Astrophysical Sciences — 2022

University of Cambridge, Cambridge, Cambridgeshire, UK

MPhil in Archaeology of the Americas — 2019 – 2020

Dissertation Title: "New Approaches to the Archaeoastronomy of Chaco Canyon"

Harvard University, Cambridge, Massachusetts, USA

Cum Laude Joint A.B. in Astrophysics-Physics — 2015 – 2019

Secondary in Archaeology

PROFESSIONAL APPOINTMENTS

NSF Astronomy & Astrophysics Postdoctoral Fellow — Sep 2026 - *Present*
Center for Astrophysics | Harvard & Smithsonian

AGEL Postdoctoral Fellow — Aug 2025 - Aug 2026
Center for Astrophysics | Harvard & Smithsonian

Resident Tutor
Currier House, Harvard University — 2025 – *Present*

Graduate Fellow
Liechtenstein Institute on Self Determination (LISD), Princeton University — 2023 – 2025
Working with Prof. Danspeckgruber

Non Resident Tutor
Currier House, Harvard University — 2022 – 2025

Graduate Assistant in Instruction
Department of Astrophysical Sciences, Princeton University
AST 203 - The Universe (Spring 2022) | AST 205 - Planets of the Universe (Fall 2021)

Resident Graduate Student
Forbes College, Princeton University — 2021 – 2025

Graduate Researcher
Department of Astrophysical Sciences, Princeton University — 2020 – 2025

Graduate Teaching Fellow
Princeton University Preparatory Program (PUPP) — 2020 – 2023

Gates Cambridge Scholar
University of Cambridge — 2019 – 2020

NASA/JPL Intern
Caltech SURF Program, Jet Propulsion Laboratory, California Institute of Technology — 2018

SAO Intern
Event Horizon Telescope, Smithsonian Astrophysical Observatory, Harvard University — 2017 – 2018

Clay Telescope Observer - Group Leader
Harvard Observing Project – Center for Astrophysics | Harvard & Smithsonian — 2016 – 2019

Affiliated Student - Banneker Institute
Center for Astrophysics | Harvard & Smithsonian — 2016 – 2017

Tutor - Bureau of Study Counsel
Harvard University 2016 – 2019

AWARDS

FELLOWSHIPS & SCHOLARSHIPS

NSF Astronomy & Astrophysics Postdoctoral Fellowship 2026
Vera Rubin Visiting Postdoctoral Fellow, Dept. of Astronomy & Astrophysics, UC Santa Cruz 2026
Ford Foundation Pre-Doctoral Fellowship, National Academies of Sciences, Engineering & Medicine 2022
Princeton Graduate Scholars Program (GSP) 2020
President's Fellowship - Princeton University Graduate School 2020
Gates Cambridge Scholarship 2019
Caltech Summer Undergraduate Research Fellowship 2018
Harvard College Research Program Summer Research Fellowship 2017 & 2016

GRANTS

Hubble Space Telescope Cycle 34 Program, PI (20 Orbits, $123,900) 2026
Nancy Grace Roman Space Telescope Cycle 1 Program 19065, PI ($450,000) 2026
NASA ADAP (Chandra Source Catalog), Co-I (PI: M. Azadi) 2026
Smithsonian Institution Asian Pacific American Initiative Pool ($35,000) 2026
NSF Astronomy & Astrophysics Postdoctoral Fellowship ($330,000) 2026

STUDENT AWARDS

Graduate Student Service Award, Association of Princeton Graduate Alumni 2025
Mentorship Award, Best of Access, Diversity and Inclusion Awards - Princeton University 2023
Group Programming Award, Best of Access, Diversity and Inclusion Awards - Princeton University 2023
Equity Award, Department of Astrophysical Sciences, Princeton University 2023
2020 Breakthrough Prize in Fundamental Physics Co-Recipient (EHT) 2019
Eagle Scout 2012

OBSERVING EXPERIENCE AND PROPOSALS

To date, I have been PI of 2 accepted *Hubble Space Telescope* proposals (23 orbits) and 1 Magellan/LLAMAS proposal, and PI of an accepted *Roman* Cycle 1 analysis program. A list of accepted PI proposals follows:

3. *AGEL020613-011417: an Einstein Spiral Lens and a New Cosmic Scale for Weighing Super-Massive Black Holes* 09/2026
   3 orbits; *HST* Cycle 34 DDT; ID: 18479
2. *Illuminating the Dark Universe with Strong Gravitational Lensing* 07/2026
   20 orbits; *HST* Cycle 34 GO (Hubble–Roman Science); ID: GO-18442
1. *Cosmology and Galaxy Evolution with Strong Gravitational Lenses* 12/2025
   0.5 night; Magellan Baade/LLAMAS 2026A (CfA TAC); ID: 802

SELECTED PUBLICATIONS

A full list of publications can be found here.

PAPERS

Córdova Rosado, R., Goulding, A., Greene, J., et al. "Cross-correlation of Luminous Red Galaxies with ML-selected Active Galactic Nuclei in HSC-SSP. III. HOD Parameters for Type I and Type II Quasars", *The Astrophysical Journal*. Mar 2026. DOI:10.3847/1538-4357/ae4230

Gottemoller, W., Sahu, N., **Córdova Rosado, R.,** Iwamoto, L., Watson, C., Tran, K.V., et al."Most strong lensing deflectors in the AGEL survey are in group and cluster environments", *The Astronomical Journal.* Mar 2026 DOI: 10.3847/1538-3881/ae4480

Córdova Rosado, R., Goulding, A., Greene, J., et al. "Cross-correlation of Luminous Red Galaxies with ML-selected Active Galactic Nuclei in HSC-SSP. II. AGN classification and clustering with DESI spectroscopy", *The Astrophysical Journal*. Dec 2025. DOI: 10.3847/1538-4357/ae182e

Shajib, A., **et al.** "A supermassive black hole 7.1 Gyr ago in the first Einstein Spiral gravitational lens", *Submitted*.

Córdova Rosado, R., Goulding, A., Greene, J., et al. "Cross-correlation of Luminous Red Galaxies with ML-selected Active Galactic Nuclei in HSC-SSP: Unobscured AGN residing in more massive halos", *The Astrophysical Journal*, Dec 2024. DOI: 10.3847/1538-4357/ad8342

Danieli, S., **et al.** "Merian: A Wide-Field Imaging Survey of Dwarf Galaxies at z 0.06-0.10 ", *Submitted to The Astrophysical Journal*, Oct 2024. DOI: 10.48550/arXiv.2410.01884

Córdova Rosado, R., Santini, L. M., "The Solstitial Axis: Solar Alignments in Ancient Mesoamerica", *Submitted to The Native American and Indigenous Studies Journal*.

Luo, Y., **et al.**, "The Merian survey: design, construction, and characterization of a filter set optimized to find dwarf galaxies and measure their dark matter halo properties with weak lensing", *Monthly Notices of the Royal Astronomical Society*, Mar 2024. DOI: 10.1093/mnras/stae925

Córdova Rosado, R., Hensley, B., Clark, S., Duivenvoorden, A., et al. "The Atacama Cosmology Telescope: Galactic Dust Structure and the Cosmic PAH Background in Cross-correlation with WISE", *The Astrophysical Journal*, Jan 2024. DOI: 10.3847/1538-4357/ad05cd

Raymond, A.; Palumbo, D.; Paine, S.; Blackburn, L. **Córdova Rosado, R.**; Doeleman, S; et al., "Evaluation of New Submillimeter VLBI Sites for the Event Horizon Telescope," *The Astrophysical Journal, Supplemental Series* Feb 2021. DOI: 10.3847/1538-3881/abc3c3

Event Horizon Telescope Collaboration **et al.**, "First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole," *The Astrophysical Journal Letters* Apr 2019. DOI:10.3847/2041-8213/ab0ec7

Event Horizon Telescope Collaboration **et al.**, "First M87 Event Horizon Telescope Results. II. Array and Instrumentation," *The Astrophysical Journal Letters* Apr 2019. DOI:10.3847/2041-8213/ab0c96

CONFERENCE POSTERS

Córdova Rosado, R., Goulding, A., Greene, J., et al.; "Stronger Clustering in Unobscured AGN with an HSC × WISE Selection" at the *2024 Ford Fellows Conference*, Washington, D.C., USA, Jun 2024.

Córdova Rosado, R., Masters, D., Rhodes, J., "Constraints on the Dark Energy Survey's Photometric Galaxy Redshifts," at the *233rd American Astronomical Society Meeting*, Seattle, Washington, USA, Jan 2019. Online Poster: https://goo.gl/1ZosGD

Córdova Rosado, R., et al., "Joint Meteorological Statistics of Observing Sites for the Event Horizon Telescope," at the *231st American Astronomical Society Meeting*, Washington, D.C., USA, Jan 2018. Press Discussion: https://goo.gl/awSb7S

PUBLIC-FACING WRITING

See translations and full text here.

Córdova Rosado, R. "Huye Luna, Luna, Luna", Jul 2023, Article link.

Córdova Rosado, R. "Homo naledi, seres de la estrella creciente", Jun 2023, Article link.

Córdova Rosado, R. "Cuatro minutos: Starship y el inicio de una nueva era de exploración espacial", *El Nuevo Día*, Apr 2023, Article link.

Córdova Rosado, R. "Universos paralelos de dos telescopios boricuas", *El Nuevo Día*, Oct 2022, Article link.

Córdova Rosado, R. "El telescopio espacial James Webb y el futuro de la astronomía puertorriqueña", *El Nuevo Día*, Jul 2022, Article link.

SERVICE & LEADERSHIP

Responsible Siting Working Group, Black Hole Initiative, Harvard University
Co-Lead — 2025 – *Present*
Member — 2024 – *Present*

AAS 249 Special Session: "Responsible Telescope Siting, Past and Future", Phoenix, AZ
Session Organizer — Jan 2027

Responsible Telescope Siting Workshop, Center for Astrophysics | Harvard & Smithsonian
Co-Organizer (with A. Thresher & S. Abelson; Heising-Simons Foundation) — Aug 2026

Culturally Sensitive Sites Working Group, IAU Division C
Member — 2026 – *Present*

Postdoc Council, Center for Astrophysics | Harvard & Smithsonian
Member 2026 – *Present*
Postdoc Symposium Organizer Oct 2026

UMBRELA Dialogue Series, Center for Astrophysics | Harvard & Smithsonian
Organizer 2025 – *Present*

Native Alumni of Princeton, Princeton University Alumni Association
Co-Leader 2026 – *Present*

NSF Review Panel
Panelist 2026

Honors Committee, Society of Indigenous Physicists
Committee Member 2024 – *Present*

Confronting simulations with the real Universe, Napa CA
Scientific Organizing Committee 2026

Princeton American Indian & Indigenous Studies Working Group, Princeton University
Organizing Board 2023 – 2025
Co-Coordinator 2024 – 2025

Native Graduate Students of Princeton, Princeton University
Organizing Board 2022 – 2025
President 2024 – 2025

Public Observing Program, Princeton Department of Astrophysical Sciences
Telescope Trainer 2022 – *2025*
Co-Coordinator 2022

National Program Subcommittee, Order of the Arrow, Boy Scouts of America
Board Member 2021 – 2024

The Peoples' Sky Conference, Council of Science and Technology, Princeton University
Co-Lead Organizer 2023 – 2024

National Scholarship Committee, 2023 National Jamboree, Boy Scouts of America
Director 2022 – 2023

Climate Committee, Princeton Department of Astrophysical Sciences
Graduate Student Representative 2021 – 2023

Graduate Student Committee, Princeton Department of Astrophysical Sciences
Graduate Student Representative 2020 –2022

Advisory Group to the Diversity and Inclusion National Committee, Boy Scouts of America
Founding Member 2020 – 2023

Camp Guajataka Summer Program, San Sebastián, Puerto Rico
Archaeology Instructor 2020

Gates Scholars' Sustainability Working Group, University of Cambridge
External Action Group Leader 2019 – 2020

Student Astronomers of Harvard-Radcliffe (STAHR), Harvard University
President & Observatory Director 2018 – 2019

Harvard College Astrophysical Society, Harvard University
Co-President 2017 – 2018

Harvard Archery Team, Harvard University
President/ USA Archery Level 2 Instructor 2016 – 2018

PRESENTATIONS

INVITED TALKS

Guest Speaker
Swiss Study Foundation Summer Academy, "Artificial Super Intelligence", Magliaso, Switzerland Sep 2026

Hernquist Research Group
Center for Astrophysics | Harvard & Smithsonian Aug 2026

Panel Presentation: What's Up, and What's Going Down, with the Skies?
Native American and Indigenous Studies Association Annual Meeting, Temuco, Chile Jul 2026

Eisenstein Research Group
Center for Astrophysics | Harvard & Smithsonian Jun 2026

FLASH Talk
Department of Astronomy & Astrophysics, University of California, Santa Cruz Feb 2026

Astronomy & Astrophysics Colloquium
Department of Astronomy & Astrophysics, University of California, Santa Cruz Feb 2026

Conference Presentation
Confronting simulations with the real Universe, Napa, California Jan 2026

Astronomy Seminar
Department of Physics & Astronomy, Dartmouth College Dec 2025

UMBRELA Dialogue Presentation
Center for Astrophysics | Harvard & Smithsonian Oct 2025

Suyu Research Group
Max Planck Institute for Astrophysics – Garching Oct 2025

Hernquist Research Group
Center for Astrophysics | Harvard & Smithsonian Oct 2025

Galaxy Group Lunch
Steward Observatory – University of Arizona Astronomy Department Nov 2024

KIPAC Tea Talk
Kavli Center for Astrophysics and Cosmology, Stanford University Nov 2024

Friday Lunch Seminar
Black Hole Initiative, Harvard University Oct 2024

ITC Luncheon
Center for Astrophysics | Harvard & Smithsonian Oct 2024

Galaxy Lunch
Astronomy Department, Yale University Oct 2024

Monday Afternoon Talk
MIT Kavli Institute for Astrophysics and Space Research Oct 2024

Oral Presentation
Seminar in honor of HSH, The Hereditary Prince Alois of Liechtenstein & HRH, The Hereditary Princess Sophie of Liechtenstein, LISD, Princeton University Sep 2023

Gravity Group Presentation
Department of Physics, Princeton University Oct 2023

CONTRIBUTED TALKS AND OTHER ACADEMIC ENGAGEMENTS

Invited Participant
Oceti Sakowin Storymap Gathering Symposium, Jeffers Petroglyphs, Minnesota Oct 2026

Contributed Talk
Inaugural CfA Science Meeting, Center for Astrophysics | Harvard & Smithsonian Aug 2026

CfA-Wide Presentation: OIR Division & AGEL (with K.-V. Tran)
Center for Astrophysics | Harvard & Smithsonian Jun 2026

Panelist
ngEHT History, Philosophy & Culture Day May 2026

Flash Talk
PASTA Postdoc Symposium, Center for Astrophysics | Harvard & Smithsonian May 2026

Panelist: Agentic AI in Science
Harvard University (organized by C. Stubbs) Apr 2026

Oral Presentation
Gravitational Lensing at the Turning Point: Current Discoveries, New Frontiers, Sexten Center for Astrophysics, Sexten, Italy Jul 2026

Dissertation Talk:
Black holes and dark matter: new insights from wide-field cosmological surveys
247th *Meeting of the American Astronomical Society, Phoenix, AZ* Jan 2026

Paper Presentation: ***The Solstitial Axis: Solar Alignments in Ancient Mesoamerica***
Native American and Indigenous Studies Association Annual Conference, Oklahoma City, OK Jun 2025

Oral Presentation: ***Black hole growth and connections to dark matter halos: new insights from wide-field cosmological surveys***
245th *Meeting of the American Astronomical Society, National Harbor, MD* Jan 2025

Oral Presentation: ***Teotihuacan, Quasars, and Indigenous Astronomy in the 21st Century***
The Peoples' Sky Symposium, Princeton University Nov 2024

Oral Presentation
The Origin and Evolution of Supermassive Black Holes Conference, Sexten Center for Astrophysics, Sexten, Italy Jul 2024

Oral Presentation
Asterisms: The Relations among their Verbal, Numerical, and Visual Representations Across Cultures in Research and Public Outreach, Institute for Advanced Study, Princeton Feb 2024

Panel Discussion after screening of *Killers of the Flower Moon*
Princeton University Nov 2023

CMB Group Meeting Presentation
Astronomy Department, Harvard University Jul 2023

Oral Presentation
From the Galaxy to the Big Bang Conference, Bayuls Sur Mer, France Jun 2023

Oral Presentation
CMB Foregrounds and Galactic Science Workshop, Tenerife, Spain Dec 2022

Panel Discussion with Osage Nation Principal Chief, Geoffrey Standing Bear
Department of Anthropology, Princeton University Nov 2022

Collaboration Meeting Presentation
Atacama Cosmology Telescope Collaboration Oct 2022

Collaboration Meeting Presentation
Atacama Cosmology Telescope Collaboration Jul 2021

Dissertation Presentation
Department of Archaeology, University of Cambridge Mar 2020

Senior Thesis Presentation
Center for Astrophysics | Harvard & Smithsonian Apr 2019

JPL Astrophysics Summer Intern Colloquium
Jet Propulsion Laboratory, California Institute of Technology Aug 2018

GUEST LECTURES

Guest Lecture
Philosophy of Science (PHIL 4510, Prof. A. Thresher), Northeastern University Oct 2026

Should LLMs Do Science? / Can We Trust LLM Science? (2 Tutorial Series)
SAORA (SPARK post-baccalaureate, Southampton & SAO predoctoral researchers), CfA Oct 2026

Should LLMs Do Science? / Can We Trust LLM Science? (2 Tutorial Series)
CASRA Summer REU Program, CfA Jun 2026

Panel on Organizing Research Notes and Files (with J. McClung)
CASRA Summer REU Program, CfA Jun 2026

Archaeoastronomy of North America and Ancient Mexico (2 Lecture Series)
An Overview of Ancient Astronomy Around the World (FRS 155), Princeton University Nov 2024, 2025

Osage History, Literature, Past, Present, and Future
Native American Literature: An Introduction (ETHN 0090D), Brown University Oct 2024

Solstices and Hilltops: A Brief Overview of Indigenous Astronomy
Native American History (HIS 271 / AMS 271), Princeton University Sep 2023

PUBLIC TALKS

Ongoing series on Astronomy, Physics, and Archaeoastronomy
Daposka Ahnkodapi (Our School), Osage Nation, Pawhuska, OK 2022 – *Present*
Recent review articles: 2024, 2025

Panel: Early Science with the *Roman Space Telescope*
Museum of Science, Boston Aug 2026

Explore the Universe: Featuring Astrophysicist Rodrigo Córdova Rosado, a Native American Heritage Month Event
Museum of Science, Boston Nov 2024

The History of the Universe, as Told by Dark Matter
TEDxMahwah, Mahwah, New Jersey Aug 2024
Video accessible here.

From Puerto Rico to the Study of Quasars
Hispanics Inspiring Students' Performance and Achievement (HISPA) Kick-off Event Sep 2023

Escuchando el vacío: La búsqueda de vida extraterrestre y Arecibo
Arecibo Observatory, Puerto Rico Oct 2022

Los agujeros negros: qué son y cómo los observamos
Arecibo Observatory, Puerto Rico May 2022

The History of the Universe
STEAM Teaching at Arecibo Academy, Arecibo Observatory, Puerto Rico Apr 2022

Astronomía y la Arqueoastronomía
*STEAM Teaching at Arecibo Academy, Arecibo Observatory, Puerto Rico*Mar 2021, Mar 2022, Oct 2022

Cosmología y Agujeros Negros
Colegio San Antonio & Colegio San Ignacio, San Juan, Puerto Rico Aug 2022

La Historia de Día de los Muertos
Day of the Dead Event, Office of Access, Diversity & Inclusion, Princeton University Nov 2021

Astronomía como Carrera Científica
Centro CRECE, San Juan, Puerto Rico Nov 2021

Cosmología Moderna
STEAM Teaching at Arecibo (STAR) Academy, Arecibo Observatory, Puerto Rico May 2021

Archaeoastronomy of the American Southwest
Internal Symposium, Sidney Sussex College, Cambridge, UK Feb 2020

The Archaeoastronomy of Chaco Canyon
Internal Symposium, Gates Cambridge Scholars, University of Cambridge Feb 2020

La Primera Sombra de un Hoyo Negro
Colegio San Ignacio de Loyola, San Juan, Puerto Rico Sep 2019

Public Observing Night with PBHA Mission Hill Afterschool Program
Loomis-Michael Observatory, Harvard University Mar 2019

Eclipse Solar 2017
Colegio San Ignacio de Loyola, San Juan, Puerto Rico Aug 2017

PROFESSIONAL ASSOCIATIONS

American Physical Society (APS), Division of Astrophysics (DAP)

American Astronomical Society (AAS)

Society of Indigenous Physicists (SIP)

Native American and Indigenous Studies Association (NAISA)

LANGUAGES

Spanish & English: Bilingual.

Osage: Intermediate.

Ancient Greek: Intermediate (reading); basic (speaking, writing).

SKILLS

Python, Linux/Unix, LaTeX, MATLAB, Mathematica, C, Adobe Lightroom, American Mountain Guide Association (AMGA) Certified Climbing Instructor